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STANDARD OPERATING PROCEDURE

Glucose Testing using Cobas c 111

Field	Detail
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Approval and Authorisation

Role	Name / Designation	Signature	Date
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Reviewed by	**Kelvin SAMAHANI** **Laboratory Manager**		
Authorized by	**Dr. Déogratias KANEZA** **Hospital's Director General**		

Review History

Prepared/Reviewed by	Version	Date	Description of Change
N/A	001	15/05/2019	First / original version
Elise NIMUKUZE	002	15/04/2021	Version and effective date update
Aurélie YANSONEYE	003	15/03/2023	Version and effective date update
Kelvin SAMAHANI	004	16/03/2025	Logo, version, effective date
Kelvin SAMAHANI	005	14/11/2026	Full revision to the 2026 standard layout aligned with ISO 15189:2022: specimen handling, calibration and traceability, two-level control with Westgard rules, measuring range, interferences, reference intervals in both units, clinical decision limits, critical values, interpretive notes, risks and records.

Controlled Copies

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1. Introduction

Glucose is the body's main source of energy. Its concentration in blood is kept within narrow limits by uptake from the gut, storage and release by the liver, and use by muscle and other tissues. Glucose is measured to diagnose and monitor diabetes mellitus, to detect hypoglycaemia, and in pancreatic, metabolic, endocrine and critical illness.

Ruli District Hospital Laboratory measures glucose in serum and plasma on the Roche cobas c 111 analyser with the Glucose HK (GLUC2) reagent. Requests and results are kept in the EMR.

2. Principle of the Test

The GLUC2 assay is an enzymatic ultraviolet test based on the hexokinase reference method. Hexokinase phosphorylates glucose to glucose-6-phosphate with ATP; glucose-6-phosphate dehydrogenase then oxidises it with NADP+ to gluconate-6-phosphate. The rate of NADPH formation is directly proportional to the glucose concentration and is measured photometrically at 340 nm. The method is standardised by the manufacturer against isotope dilution mass spectrometry (ID-MS).

3. Purpose

This SOP standardises the pre-examination, examination and post-examination steps for the quantitative measurement of glucose in serum and plasma on the cobas c 111, so that results are accurate, reproducible and clinically reliable.

4. Scope

This SOP applies to all laboratory biotechnologists who operate the cobas c 111 and perform glucose testing in the biochemistry section. It covers specimen receipt, calibration, quality control, analysis, validation and reporting. It does not cover glucose measured with a glucometer at the bedside (point-of-care), or glucose in urine or cerebrospinal fluid.

5. Definitions and Abbreviations

5.1 Definitions

No.	Term	Definition
1	Fasting glucose	Glucose measured after at least 8 hours without food or drink other than water.
2	Random glucose	Glucose measured at any time, without regard to the last meal; interpreted with the collection time and the patient's state.
3	Glycolysis	Use of glucose by blood cells in the unseparated tube, which lowers the measured value (about 7% per hour at room temperature).
4	Calibration	Setting the relationship between the instrument signal and the glucose concentration with a calibrator of known value.
5	Measuring range	The interval of concentrations the method reports without dilution.
6	Westgard rules	Statistical rules used to decide whether a control result is acceptable.

5.2 Abbreviations

No.	Acronym	Full Meaning
1	C.f.a.s.	Calibrator for automated systems (Roche)
2	CV	Coefficient of Variation
3	EHO	Environmental Health Officer
4	EMR	Electronic Medical Record
5	EQA	External Quality Assessment
6	GLUC2	Roche Glucose HK reagent, second generation
7	IPPIS	Integrated Personnel and Payroll Information System
8	IQC	Internal Quality Control
9	LIS	Laboratory Information System

No.	Acronym	Full Meaning
10	NRL	National Reference Laboratory (RBC)
11	PPE	Personal Protective Equipment
12	RBC	Rwanda Biomedical Centre
13	RliDHL	Ruli District Hospital Laboratory
14	SD	Standard Deviation
15	SOP	Standard Operating Procedure
16	TAT	Turnaround Time

6. Specimen Requirements

6.1 Request and patient identification

- Accept requests entered in the EMR by an authorised requester. The request states the patient's identifiers, the examination, and whether the specimen is fasting, random or timed — the result cannot be interpreted without this.
- Identify the patient with at least two identifiers matched to the EMR request; label the specimen in the patient's presence (Pre-Examination Procedure, RliDHL-ADM-SOP-005).

6.2 Patient preparation and collection

- Fasting glucose: no food or drink except water for at least 8 hours; record the fasting state and the exact collection time in the EMR.
- Do not collect from an arm into which a glucose infusion is running.
- Collect venous blood according to Venous Blood Collection (RliDHL-IH-SOP-001): a plain or clot-activator tube for serum, or a fluoride/oxalate tube when the specimen cannot reach the laboratory and be centrifuged within 30 minutes.

6.3 Specimen types, volume, and storage

Specimen	Minimum volume	Handling and storage
Serum (plain or clot-activator tube)	300 µL in the sample cup	Centrifuge and separate within 30 min; separated serum: 8 h at 15–25 °C, 72 h at 2–8 °C
Plasma (lithium heparin or EDTA)	300 µL in the sample cup	As for serum
Fluoride/oxalate plasma	300 µL in the sample cup	Glucose preserved in the unseparated tube: at least 24 h at 15–25 °C

NOTE: in blood left unseparated at room temperature, glucose falls by about 7% in one hour. A specimen in a tube without fluoride that reaches the laboratory late gives a falsely low result.

6.4 Specimen acceptance and rejection criteria

Criterion	Acceptable	Reject / request recollection
Labelling	Fully labelled; matches the request; collection time and fasting state given	Unlabelled, mislabelled, or mismatch with the request
Container	Plain, clot-activator, heparin, EDTA or fluoride/oxalate tube	Wrong tube; leaking or broken tube
Volume	At least 300 µL of serum or plasma	Insufficient volume
Quality	Clear serum or plasma; slight haemolysis, icterus or lipaemia accepted and noted on the report	Specimen drawn from an infusion arm; grossly contaminated specimen
Age	Separated within 30 min, or collected in fluoride; within the storage limits above	Unseparated, non-fluoride blood older than 30 min; beyond the limits

Record every rejected specimen in the specimen rejection register and inform the requester at once. If a specimen that fails these criteria is irreplaceable or clinically critical, the Laboratory Manager or designee may authorise testing; the report then states the caution needed to interpret the result.

7. Responsibilities and Authorisation

Task	Responsible Person(s)
Authorise this SOP and provide the resources needed to apply it	Hospital Director General — Dr. Déogratias KANEZA
Train, assess and authorise staff on this SOP; review it at least every two years; validate and release results	Laboratory Manager — Kelvin SAMAHANI
Monitor IQC, EQA, nonconformities and corrective actions for this procedure; keep the master list and controlled copies up to date	Lab Quality Manager — Jean Paul NSENGIYUMVA
Ensure reagents, controls and consumables are in stock, within expiry and stored correctly	Laboratory Manager / Hospital Pharmacist
Perform start-up checks, calibration, quality control and glucose testing on the cobas c 111; record results	Authorised Laboratory Biotechnologists
Perform daily, weekly and monthly maintenance of the cobas c 111 and record it	Laboratory Biotechnologist on duty
Dispose of biological and chemical waste safely; supervise waste segregation	Environmental Health Officer (EHO); all laboratory staff

Authorisation. Only staff trained on this SOP, assessed as competent and authorised by the Laboratory Manager perform this procedure and release its results without supervision. Staff in training work under the direct supervision of an authorised person, who checks and countersigns their work. Training, competence assessment and authorisation are recorded in the personnel training file; acknowledgement of this SOP is recorded on the Signature Page (section 21).

8. Equipment, Reagents, and Consumables

8.1 Equipment

- Roche cobas c 111 analyser; serial number and asset tag in the instrument logbook.
- Uninterruptible power supply (UPS).
- Centrifuge; refrigerator at 2–8 °C for reagents, calibrator and controls (temperature recorded twice daily).
- Micropipettes (20–200 µL and 100–1000 µL).

8.2 Reagents, calibrators and controls

- Glucose HK (GLUC2) reagent for the cobas c 111; store at 2–8 °C. Record the lot number, expiry date and loading date; respect the on-board stability in the method sheet.
- Calibrator: C.f.a.s., reconstituted and stored as its insert directs; deionised water is the zero calibrator.
- Controls: PreciControl ClinChem Multi 1 (PCC1, normal) and Multi 2 (PCC2, pathological), or equivalent with assigned values. Follow the insert for reconstitution and stability.
- On receipt, check each consignment for damage, transport conditions, lot number and expiry; enter it in the stock records.
- When a new reagent lot is loaded, calibrate and then run both control levels before patient specimens. When a new control lot is started, run it alongside the lot in use to confirm its target values.
- Do not use any reagent, control or device past its expiry date or if storage conditions were not met. Keep the manufacturer’s current instructions for use at the bench.
- Report any defect or adverse incident linked to a reagent, kit or instrument to the Laboratory Manager, who records it and informs the supplier and, where required, RBC/NRL and the Rwanda Food and Drugs Authority.

8.3 Consumables

- Sample cups, cuvette segments, pipette tips, tube racks.
- Gloves, laboratory coat, biohazard and liquid waste containers, 10% sodium hypochlorite, 70% ethanol.

8.4 Calibration and metrological traceability

- Calibration mode: linear, two points: deionised water and C.f.a.s.
- Calibrate after each change of reagent lot, after major maintenance, and whenever quality control shows a shift or failure traced to calibration.
- Check that the instrument accepts the calibration, then run both control levels before any patient specimen. Record the calibrator lot, date and acceptance in the instrument logbook.

- Traceability: standardised by the manufacturer against isotope dilution mass spectrometry (ID-MS), the reference method for glucose.

9. Environment and Biosafety Measures

- Keep the analyser on a stable surface, away from direct sunlight, vibration, dust, air-conditioning outlets and heat sources. Record the room temperature each day.
- Reagents contain hazardous chemicals: read the safety data sheet; avoid contact with skin and eyes.
- Treat every specimen as potentially infectious: laboratory coat and gloves; eye protection where splashing is possible; hand hygiene after removing gloves.
- Clean spills with 10% sodium hypochlorite; decontaminate the bench with 70% ethanol at the end of each session.
- After any exposure, wash immediately, inform the Laboratory Manager and follow the post-exposure procedure in the Laboratory Safety Manual.
- Discard used devices, tips, tubes and specimens as infectious waste; empty the liquid waste container of the analyser as infectious liquid waste (Laboratory Waste Management procedure; supervised by the EHO).

10. Procedural Steps

10.1 Instrument start-up and daily checks

1. Check that the analyser and the UPS are connected. Switch on the cobas c 111 and wait for the self-check.
2. Check the system water and the waste container; refill or empty as needed.
3. Check the GLUC2 reagent: tests remaining, on-board expiry and lot number. Load a new bottle if needed and record it in the reagent log.
4. Confirm the reaction temperature shown by the analyser is 37 °C (± 0.5 °C). Perform the daily maintenance the instrument asks for and record it.

10.2 Calibration

Calibrate when section 8.4 requires it. Bring the reconstituted calibrator to room temperature, run the calibration from the analyser menu, and continue only when the calibration is accepted.

10.3 Internal quality control

- Run PCC1 and PCC2 like patient specimens at the start of each day of testing before any patient specimen, and after every calibration.
- Plot each result on the Levey-Jennings chart and apply the rules in section 11.
- Proceed to patient specimens only when both levels are acceptable.

10.4 Specimen preparation and loading

1. Check the identity of the specimen against the request (names, number, test, date and time of collection).
2. Centrifuge the tube at 3000–5000 rpm for 5 minutes. Inspect the serum or plasma for haemolysis, lipaemia and icterus and note what you see.
3. Label a sample cup with the laboratory code and pipette at least 300 µL of serum or plasma into it.
4. Enter or check the glucose (GLUC2) order in the worklist, place the cup in the indicated position, confirm the position matches the laboratory code, and start the run.

10.5 Analysis and result review

- The analyser performs the reaction and calculates the glucose concentration.
- Review each result on the screen for flags and alarms.
- If the result is above the measuring range (above 41.6 mmol/L / 750 mg/dL), dilute with 0.9% sodium chloride as the method sheet directs, repeat, and multiply by the dilution factor.
- If a result is implausible or not matching earlier results, check the specimen for late separation or contamination and repeat before reporting.

10.6 Post-procedure steps

- Cap and keep the specimens as described in section 16.
- Discard used cups and tips as infectious waste; decontaminate the bench.
- Perform the end-of-day maintenance and shutdown steps and record them.

11. Quality Control and External Quality Assessment

11.1 Internal quality control

- Two levels (PCC1 and PCC2) at the start of each day of testing, after each calibration, after a reagent lot change, and after maintenance or repair.
- Rules: one result outside ± 2 SD is a warning — inspect the chart and repeat the control. One result outside ± 3 SD, two consecutive results outside ± 2 SD on the same side of the mean, or a range of 4 SD between levels is a rejection.
- Plot every result on the Levey-Jennings chart. The Laboratory Manager or Lab Quality Manager reviews the charts each week for shifts and trends and each month calculates the mean, SD and CV for each level.
- Record control results, lot numbers and any action in the quality control register.

11.2 External quality assessment

The laboratory takes part in the clinical chemistry EQA scheme organised or designated by RBC/NRL and tests the samples in the same way as patient specimens. Unsatisfactory performance is investigated as in 11.3.

11.3 Actions when a control fails

Stop patient testing, identify the cause (reagent, calibration, instrument or specimen), correct it, and repeat both control levels before resuming. Assess whether patient results reported since the last acceptable control are affected; repeat affected specimens and issue corrected reports where needed (ISO 15189:2022, 7.5, 8.7). Record everything in the quality control register and the nonconformity log.

12. Result Interpretation and Validation

The analyser reports glucose in mmol/L (conversion: mg/dL ÷ 18.02 = mmol/L). Before release, the Laboratory Manager or authorised biotechnologist checks: the result against the previous result (delta check) and the clinical information; the control status of the run; the specimen notes (haemolysis, icterus, lipaemia, infusion arm, delayed separation); and the flags raised by the analyser. Results are validated in the EMR under the user's account (ISO 15189:2022, 7.4.1).

Observation	Interpretive note (common causes)
High glucose (hyperglycaemia)	Diabetes mellitus; stress response in critical illness; corticosteroids or thiazides; Cushing syndrome; pancreatitis; parenteral nutrition; specimen from a glucose infusion arm (falsely high).
Low glucose (hypoglycaemia)	Insulin or oral hypoglycaemic excess; missed meals or prolonged fasting; sepsis; liver failure; adrenal insufficiency; insulinoma; neonatal hypoglycaemia.
Falsely low result	Delayed separation of a non-fluoride tube (glycolysis, about 7% per hour); excess fluoride dilution.
Result not matching the patient	Repeat the specimen; check identification and collection notes before release; treat mislabelling as nonconforming work (7.5).

13. Performance Characteristics and Limitations

- Measuring range: 0.11–41.6 mmol/L (2–750 mg/dL); dilute and repeat above the range (10.5). Verify against the method sheet of the lot in use.
- Within-laboratory precision: CV at most 2.5% at concentrations around 5 and 15 mmol/L (see the method verification file).
- Interferences: severe haemolysis, icterus and lipaemia affect the result according to the indices given in the instructions for use; note any accepted interference on the report.
- Measurement uncertainty: see the method verification file (ISO 15189:2022, 7.3.4); where relevant it is stated with the result.

14. Reference Intervals, Clinical Decision Limits, and Critical Results

Normal range in use at this laboratory (Laboratory Normal Ranges list, Ruli District Hospital, approved 08/07/2025 by the Laboratory Manager and the Hospital Director General):

Test / specimen	Normal range (in use)	Normal range (mmol/L)
Serum glucose	70 – 110 mg/dL	3.9 – 6.1

The range is verified before first use and after any method or reagent change (ISO 15189:2022, 7.3.5). Interpret paediatric and neonatal results in the clinical context.

Clinical decision limits for the diagnosis of diabetes (WHO/ADA criteria):

Test / state	Decision limit	Meaning
Fasting plasma glucose	>= 7.0 mmol/L (>= 126 mg/dL)	Diabetes mellitus
Fasting plasma glucose	5.6 – 6.9 mmol/L (100 – 125 mg/dL)	Impaired fasting glucose
2-hour oral glucose tolerance test	>= 11.1 mmol/L (>= 200 mg/dL)	Diabetes mellitus
2-hour oral glucose tolerance test	7.8 – 11.0 mmol/L (140 – 199 mg/dL)	Impaired glucose tolerance
Random glucose with classic symptoms	>= 11.1 mmol/L (>= 200 mg/dL)	Diabetes mellitus
Hypoglycaemia alert	< 3.9 mmol/L (< 70 mg/dL)	Alert value — inform the ward
Clinically significant hypoglycaemia	< 3.0 mmol/L (< 54 mg/dL)	Treat without delay

Critical results (Laboratory Critical Results list, Ruli District Hospital): glucose below 40 mg/dL (2.2 mmol/L) or at 399.6 mg/dL (22.2 mmol/L) and above. Report every critical result immediately after verification, even if other tests in the panel are still in progress. Notification follows the mandatory ‘Write Down and Read Back’ process; record the name of the person informed and the date and time (ISO 15189:2022, 7.4.1.3).

15. Reporting and Release of Results

Turnaround time (Laboratory Tests TAT list, Ruli District Hospital): 45 minutes from specimen receipt to result communication. Results are communicated immediately after verification, even if other tests in the panel remain in progress.

- The report shows: patient identifiers, examination, result with units (mg/dL and mmol/L), the normal range in use for this laboratory, specimen notes, collection time and the reporting laboratory.
- Critical results (14) are reported immediately using the mandatory ‘Write Down and Read Back’ process, and the communication is recorded.
- Corrected reports: issue an amended report identifying the change, the reason and the date; notify the requester; retain the original entry.
- Patient information is confidential (ISO 15189:2022, 4.2).

16. Specimen Retention and Disposal

- After reporting, keep specimens for 7 days at 2–8 °C for repeats or investigation, then dispose of them under the Laboratory Waste Management procedure.
- Specimens under investigation for a nonconformity or complaint are retained until the case is closed.

17. Risk Management, Nonconforming Work, and Contingency

Risk	Control measure
Wrong-patient result released	Identification at collection (RliDHL-IH-SOP-001); barcode check at loading; delta check before release
Control failure not detected	Two-level IQC before patient testing; Westgard rules; weekly chart review
Reagent lot shift	Calibration and both control levels on every new lot before patient specimens
Power interruption	UPS; record the event; repeat controls after power restoration before testing
Critical result not communicated	Telephone with read-back; record; escalate to the Laboratory Manager if unreachable
EMR/LIS downtime	Downtime procedure: report forms completed manually; results telephoned with read-back; back-entry within 24 hours of restoration

Nonconforming work (ISO 15189:2022, 7.5): when a result, control or step does not conform — stop, report to the Laboratory Manager, assess the clinical impact on results already released, repeat or recollect, and issue corrected reports where needed. Record the event and the corrective action; evaluate its effectiveness (8.7).

18. Records

Record	Where kept	Retention
Control results and Levey-Jennings charts	Quality control register	As set in the laboratory records policy
Calibration and maintenance logs	Instrument logbook	As above
Reagent, calibrator and control lot records	Stock / reagent records	As above
Temperature records (2–8 °C refrigerator)	Temperature log	As above
EQA reports and actions	Quality file	As above
Patient results and reports	EMR / LIS (electronic)	As above
Training, competence and authorisation (section 21)	Personnel training file	Duration of employment plus the period required by national rules

Records are created at the time the activity is performed, are legible, traceable to the person or user account, and are retrievable (ISO 15189:2022, 8.4; RliDHL-ADM-SOP-004).

19. Supporting Documents

- Document and Record Control (RliDHL-ADM-SOP-004).
- Pre-Examination Procedure (RliDHL-ADM-SOP-005).
- Venous Blood Collection (RliDHL-IH-SOP-001).
- Equipment Management and Maintenance procedure.
- Internal Quality Control and EQA procedure.
- Laboratory Normal Ranges list (approved 08/07/2025).
- Laboratory Critical Results list.
- Laboratory Tests TAT list (2026).
- Result Reporting and Release procedure.
- Laboratory Safety Manual; Laboratory Waste Management procedure.

20. References

- ISO 15189:2022 — Medical laboratories: requirements for quality and competence (in particular 7.3.2 to 7.3.5, 7.3.7, 7.4.1).
- Roche cobas c 111 GLUC2 method sheet and instructions for use (of the lot in use).
- WHO/ADA diagnostic criteria for diabetes mellitus and hypoglycaemia.
- CLSI GP41 — Collection of Diagnostic Venous Blood Specimens.
- WHO — Laboratory Quality Management System (LQMS) Handbook.

21. Signature Page — Training and Competence Acknowledgement (Form F-TRN-001)

I confirm that I have been trained on this SOP, that I have read and understood it, including the safety precautions, and that I agree to perform it as written. I will not perform it without supervision until this record is complete.

S/No	Full Name (print)	IPPIS No.	Designation	Signature	Date
1					
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Trainer / Supervisor: _____ Name (print): _____ Date: _____

Enter the IPPIS number (staff cards are not numbered). If more rows are needed, attach continuation sheet F-TRN-001-2.

22. Amendment Page

Handwritten amendments are authorised by the Laboratory Manager and are entered here word for word before the next revision (ISO 15189:2022, 8.3).

Date	Section	Amendment (word for word)	Reason	Authorised by